

Chemotherapy and Cancer Stem Cells

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Cancer stem cells are subsets of tumor cells defined through the capacity to initiate tumors and have been linked to therapeutic resistance. In this issue of *Cell Stem Cell*, Todaro et al. (2007) demonstrate that targeting tumor IL-4 sensitizes colon cancer stem cells to chemotherapy.

Solid tumors consist of not only neoplastic cells but also supporting vasculature, inflammatory cells, and fibroblasts. Essentially, solid tumors mimic aberrantly developed organs or tissues (Reya et al., 2001). Tumor cell populations are commonly heterogeneous with hierarchies of cellular populations that display a range of differentiation phenotypes, including specific morphologies and lineage markers. The majority of cells in bulk tumors may be nontumorigenic end cells derived from stem and progenitor cells, but subsets of tumor cells—called cancer stem cells (CSCs) or tumor-initiating cells—with proposed roles in tumor propagation have been identified in several cancer types.

The requirements for identification of a CSC are currently functional: extensive self-renewal and tumor initiation. The laboratories of John Dick and Michael Clarke laid the foundation for the CSC field by applying techniques and differentiation markers developed in normal stem cell biology to derive tumor populations enriched for cancer stem cells in leukemias and solid tumors (reviewed in Dalerba et al., 2007a). Two recent reports used the Prominin-1, CD133, cell-surface marker to enrich for CSCs in human colon cancers (O'Brien et al., 2007; Ricci-Vitiani et al., 2007), whereas the Clarke laboratory used CD44 and epithelial cell adhesion molecule (EpCAM) (Dalerba et al., 2007b). In this issue of *Cell Stem Cell*, Todaro et al. employed the CD133 marker to enrich for tumor cells derived from colon cancer surgi-

cal specimens that form spheres in serum-free culture and tumors when xenotransplanted into immunocompromised mice (Todaro et al., 2007). Of note, the CD133⁺ population represented 0.3%–3% of tumor specimens, but only 10 of 21 tumors yielded tumor spheres and secondary tumors, more commonly in tumors with microsatellite instability. Depletion of the CD133-expressing cells attenuated the tumorigenic potential, further implicating the CD133⁺ fraction in tumor initiation.

Despite the development of molecularly targeted therapies, radiation and cytotoxic chemotherapies remain mainstays of cancer treatment. Although unusual cancers like testicular cancers can be cured at advanced stage with conventional therapy, therapeutic resistance is common among most advanced cancers. Many mechanisms may contribute to the development of therapeutic resistance, including the stochastic selection of resistant genetic subclones, microenvironmental factors (hypoxia, acidosis, etc.), and cell extrinsic factors. The development of the CSC hypothesis has led to the expectation that tumor-initiating cells may display resistance to cytotoxic cancer therapies, permitting the repopulation of tumors treated with radiation or chemotherapy. Indeed, several groups, including our own laboratory, have demonstrated that CSCs or cells expressing stem cell markers from multiple cancer types exhibit resistance to conventional cancer therapies. We demonstrated that CSCs derived

from human glioblastoma surgical biopsy specimens and xenografts are resistant to the effects of ionizing radiation due to preferential activation of the DNA damage checkpoint response (Bao et al., 2006). Additional studies utilizing breast cancer cell lines confirmed that cells expressing CSC markers are resistant to radiation and demonstrated that ectopic expression of Wnt or β -catenin augmented the radioresistance of this subpopulation (Woodward et al., 2007; Phillips et al., 2006). Studies of tumor cells have also revealed that CSCs are enriched in populations that exclude Hoechst 33342, i.e., the side population (SP). Dye exclusion is mediated by efflux pumps, which may also function to exclude drugs, resulting in the observed chemoresistance of SP cells. Furthermore, leukemic CSCs display chemotherapy resistance relative to the majority of the tumor cells that are more highly differentiated (Wulf et al., 2001).

Against this background, Todaro et al. (2007) analyzed the chemosensitivity of CD133⁺ and spheroid-initiating colon cancer cells. Oxaliplatin and 5-fluorouracil, two chemotherapies employed in standard treatment regimens for colon cancer patients, were less effective against CD133⁺ tumor cells than either bulk cells or CD133[−] tumor cells. Based on their prior work, they investigated the role of interleukin-4 (IL-4) in mediating chemoresistance. Both adherent cancer cells (enriched for CD133[−] cells) and tumor spheroids (enriched for CD133⁺ cells) expressed higher levels of IL-4 ligand and receptor than normal colon. An anti-IL-4

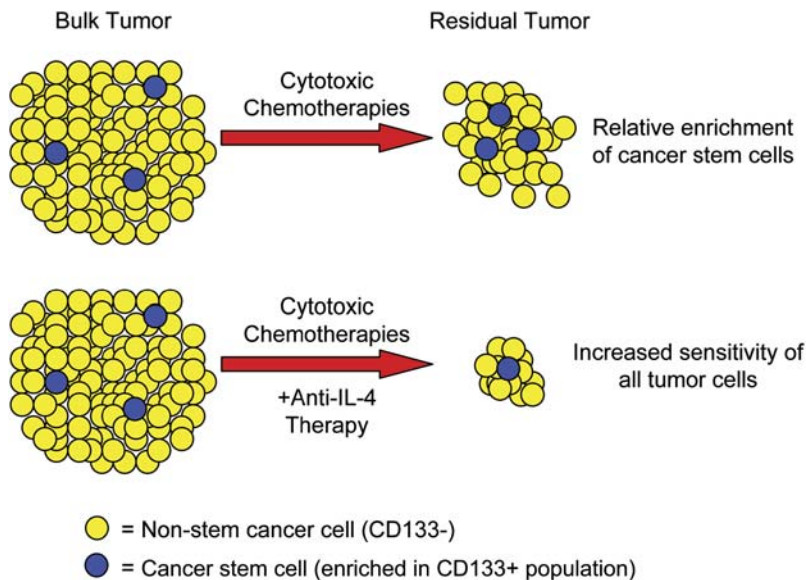


Figure 1. Cancer Stem Cells Display Chemotherapy Resistance Promoted by IL-4 Expression

Colon cancer stem cells defined through the expression of the CD133 cell-surface stem cell marker, tumor sphere formation, and tumor initiation are less sensitive to the cytotoxic effects of chemotherapy. All colon cancer cells appear to express higher levels of IL-4 than normal colon tissues, suggesting a potential therapeutic index with anti-IL-4 therapies. Targeting IL-4 activity through neutralizing antibodies or a dominant-negative IL-4 ligand increases the sensitivity of both nonstem colon cancer cells and colon cancer stem cells.

antibody reduced the viability of CD133⁻ and CD133⁺ cancer cells and increased the efficacy of chemotherapy treatment (Figure 1). The anti-IL-4 antibody induced apoptosis in colon cancer spheres associated with the reduction of prosurvival molecules. These results were validated in animal studies with heterotopic flank tumors with either a mutated, inhibitory form of IL-4 (IL-4DM) or the neutralizing IL-4 antibody. These results support a role of IL-4 in an autocrine survival signal for the entire tumor cell population that contributes to chemoresistance.

These results have important implications. First, they validate the observations that have been made in other models that CSCs exhibit therapeutic resistance to cancer treatments currently employed. Second, they identify an additional mechanism of resistance that is shared between heterogeneous tumor cell populations. The expression of IL-4 and its receptor were not specific to the CSC population, so the IL-4 axis does not explain the differential sensitivity of the CSCs. Instead these studies suggest that molecular pathways that contribute to bulk tumor

growth can be successfully targeted to sensitize CSCs to cytotoxic therapies. Further, the current availability of anti-IL-4 therapies offers a rapid translation and clinical assessment of an improved therapeutic paradigm. The precise mechanism by which IL-4 mediates chemoresistance in tumor populations remains unclear, but the authors provide strong circumstantial evidence that IL-4 regulates key antiapoptotic mechanisms that may play a functional role. A potential confounding issue is the enrichment of tumor formation in tumors with microsatellite instability. As microsatellite instability also may predict chemosensitivity, the CSC response to chemotherapy and the role of IL-4 in this response may not hold for all cancers. However, the authors did confirm their results in a tumor that was microsatellite stable.

If recurrent tumors contain greater numbers of CSCs that are also selected for greater resistance to therapy, the lethality of advanced cancers might be better explained. However, [Todaro et al. \(2007\)](#) examine a resistance mechanism (IL-4 expression) that is active in all of the tumor cells

and, in some ways, raise as many questions as they answer. In cancers that frequently do not respond to chemotherapy (including colon and other gastrointestinal cancers) is CSC resistance important? Furthermore, how does the CSC hypothesis inform models in which resistant tumor subclones emerge during therapy? Indeed, unpublished data from our own lab suggest that CSCs can give rise to clonal subpopulations in response to selective pressures. Do the models that CSCs reside in hypoxic niches or secrete proteins that may act both in autocrine and paracrine manners impact on traditional hypotheses of cancer therapeutic resistance? Finally, are mechanisms of CSC therapeutic resistance shared across cancer types? If so, the identification of common CSC resistance mechanisms may have broad utility in targeting new therapeutics. In sum, the CSC hypothesis is not a replacement for the large body of work that has defined multiple mechanisms whereby cancers are resistant to conventional therapies, but interpreting prior hypotheses through the prism of cellular heterogeneity within tumors may better model clinical observations in some cancers. In particular, given the poor capacity of current preclinical cancer models to predict therapeutic efficacy in human subject clinical trials, it would be of profound significance should CSC models yield more accurate outcome predictions. Of course, the contribution of CSCs will first require validation in human subjects. Finally, it is important to remember that, although [Todaro and coworkers](#) examined the expression of IL-4 in normal colon specimens, specific, noncancerous stem cell populations were not assessed. Future studies will certainly address the role of each resistance mechanism in cancer and normal stem cells as an essential step in the development of targeted therapeutics.

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Quo Vadis Brain Repair? A Long Axonal Journey in the Adult CNS

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Recently in *Nature Neuroscience*, Gaillard et al. (2007) study axonal projections from embryonic cortical explants grafted into acutely damaged adult motor cortex. After attempting to rule out fusion of donor tissue with pre-existing host circuitry, the authors report robust long-distance donor axonal projections and synaptic integration into target regions appropriate for the motor system.

It is an intricate journey from the developing neuroectoderm—populated by uncommitted multipotent neural stem cells (NSCs)—to the highly specific synaptic contacts that, for example, precisely move one's finger. As development of the central nervous system (CNS) proceeds, the neural tube generates billions of neurons that are interconnected through axons, establishing trillions of synapses. In this highly complex and tightly controlled network, one neuron can establish >10,000 synapses with another neuron. It seems daunting to think that a cellular suspension of exogenous NSCs transplanted into a damaged cortex of an adult brain should be able to recapitulate such precision and complexity. And, in general, it has been quite difficult to achieve. Even under conditions where functional recovery has been observed after the implantation of stem cells into the lesioned CNS, the underlying mechanism has often turned out to be not primarily neuronal replacement but rather the stem cell-mediated protection of pre-existing neuronal networks. This phenomenon, called the “chaperone

effect” (Ourednik et al., 2002), has actually come to be viewed as “low-hanging fruit” in the stem cell field for many types of abnormalities in many organ systems (e.g., heart, muscle, pancreas, bone, in addition to CNS).

Nevertheless, neuronal replacement remains the Holy Grail of regenerative medicine, although we have all come to recognize how challenging that is. A recent paper by Gaillard et al. (2007) offers some hope that we may need not abandon that lofty goal and provides some insights into what we might need to achieve as preconditions for replacement of neural circuitry.

Endogenous and grafted neural stem/progenitor cells do have the ability to generate projection neurons in the neocortex (Snyder et al., 1997; Park et al., 2002; Chen et al., 2004; Molyneaux et al., 2007)—however, with low efficiency. Prevailing thought holds that this low incidence of neuronal replacement is rooted in the realization that NSCs in a cellular suspension, removed from their instructive and contextual cues, do not efficiently acquire an identity with sufficient refinement to become neurons that

“know” to restore discrete and specialized connections. Gaillard et al. (2007), however, demonstrate that, if we retain (or, presumably, provide) those contextual cues, as one does with a fetal cortical organotypic explant, one can achieve fairly robust neuronal connectivity in a complex system. The authors used a straightforward approach. They aspirated the motor cortex of 4- to 6-month-old male mice and immediately grafted into the lesion cavity embryonic motor cortex explanted from green fluorescent protein (GFP)-expressing female mice. Animals analyzed at 2 months postgrafting showed exuberant long-distance axonal projections and anatomical restoration. Large numbers of GFP-expressing axons were found in the appropriate cortical and subcortical target areas of the motor pathway bilaterally. (Functional recovery was, unfortunately, not studied.) To verify *de novo* extension of myelinated efferents and synaptogenesis, and to rule out fusion with host cells as an alternative mechanism, a series of experiments were performed, including axonal tracing, immunogold